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Problem Statement: Write a program in C to accept a number from the user & print the digits of the number in reverse order.

Objective:
- To understand number manipulation using loops.
- To learn the use of modulo (%) & division (/) operators.
- To implement logic for reversing digits of a number
- To improve problem-solving skills using iterative structures.

Theory: To understand number manipulation using loops, To learn the use of modulo (%) and division (/) operators. To implement logic for reversing digits of a number. To improve problem-solving skills using iterative structures.

Platform: Linux.

Algorithm:

Step 1: START.
Step 2: Declare variables n, digit.
Step 3: Read the number n from user.
Step 4: While n is not equal to 0, repeat:
  a) digit ← n modulo 10
  b) print digit.
  c) n ← n divides 10
Step 5: STOP.

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C program:

```
# include <stdio.h>
int main() {
    int n, digit;
    printf("Enter a number:");
    scanf("%d", &n);
    printf("Reverse digits:");
    while(n != 0) {
        digit = n % 10;
        printf("%d", digit);
        n = n / 10;
    }
    return 0;
}
```

Input:

The user enters an integer number

Ex. Input: Number: 4567

Output:

The program prints the digits of number in reverse order.

Ex. Output: Digits in reverse order: 7 6 5 4.

Conclusion: Thus, the program to print the digits of a number in reverse order was successfully implemented using C programming. The program demonstrates the effective use of loops, modulo, operator, and divisor operator.

FAQs:

1. Why is the modulo operator (%) used in this program?

A. a) The modulo operator (%) is used to extract the last digit of a number.

b) When a number is divided by 10, the remainder is always the last digit.

Ex: If n = 1234:

n % 10 = 4 → last digit

n / 10 = 123 → remaining number.

2) Why is division by 10 used after extracting each digit?

A. After extracting the last digit using the modulo operator, division by 10 is used to remove that last digit from the number.

Ex: If n = 1234:

n % 10 = 4

n / 10 = 123

Then process repeats:

123 % 10 = 3

123 / 10 = 12

This continues until n becomes 0.

3) What happens if the input number is zero?

A. a) If the user enters 0, the condition n != 0 is false.

b) So the while loop does not execute even once.

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c) Nothing is printed as reverse digits

4) How will the program behave for negative numbers?

A) a) If the input is negative, the loop will still run
b) The modulo operator % will return negative digits.
c) The output will contain negative values, which is not a proper reverse

5) Why is a while loop preferred for this operation?

A. a) The loop runs until number becomes 0.
b) Each iteration processes one digit.
c) The stopping condition (n != 0) is naturally linked to logic of problem.
d) A while loop is preferred because the number of digits is not known in advance.